Week 1

Q1. A single-site cloud (datacenter) typically includes:
A) Compute nodes, switches, network topology, storage nodes, and a front-end for jobs/requests
B) Only end-user laptops
C) Only a single database server
D) Only Wi-Fi routers and mobile devices
Correct: A
Justification: A datacenter includes compute, network, storage, and an access/front-end layer.

Q2. We describe today's clouds as having all EXCEPT:
A) Massive scale
B) On-demand access
C) Data-intensive nature
D) Guaranteed single-thread execution
Correct: D
Justification: Modern clouds emphasize scale, elasticity, and data-intensive workloads, not single-thread guarantees.

Q3. Which *as-a-service* model is "flexible compute + storage infrastructure", often enabled by virtualization?
A) SaaS
B) PaaS
C) IaaS
D) DaaS
Correct: C
Justification: IaaS provides virtualized infrastructure resources like compute and storage.

Q4. PaaS most closely provides:
A) Bare metal hardware access only
B) A platform/runtime framework to develop and deploy apps
C) Only office productivity software
D) Only network cables and switches
Correct: B
Justification: PaaS bundles infrastructure with a software platform for application development/deployment.

Q5. Virtualization allows:
A) One OS only per machine
B) Concurrent execution of multiple OSs on the same physical machine
C) Eliminating the need for operating systems
D) Only running applications without OS
Correct: B
Justification: Virtualization enables multiple guest OS instances to run concurrently on one host.

Q6. The "virtualization layer" is also called the:
A) Compiler
B) Hypervisor / Virtual Machine Monitor (VMM)
C) Cache controller
D) DNS resolver

Correct: B
Justification: The hypervisor/VMM manages physical hardware and presents virtual resources.

Q7. Which is true about a **Type 1 (bare-metal) hypervisor** model?
A) It runs on top of a full host OS
B) It directly manages hardware resources and runs guest VMs above it
C) It cannot support VMs
D) It is only a user-space application
Correct: B
Justification: Type 1 hypervisors sit closer to hardware and manage resources directly.

Q8. In x86 hardware **without virtualization**, the OS must execute privileged instructions in:
A) Ring 3
B) Ring 2
C) Ring 1
D) Ring 0
Correct: D
Justification: The OS must execute privileged instructions in **Ring 0**.

Q9. In hotspot mitigation for VM migration, a **black-box approach** is described as:
A) Using OS and application-level statistics
B) Fully OS- and application-agnostic
C) Requiring source-code changes to applications
D) Possible only with para-virtualized guests
Correct: B
Justification: We define black-box as **fully OS- and application-agnostic** (gray-box uses OS/app stats).

Q10. Para-virtualization typically requires modifying approximately:
A) 30% of guest OS code
B) 10% of guest OS code
C) Less than 2% of guest OS code
D) 50% of guest OS code
Correct: C
Justification: Para-virtualization needs only small guest changes (noted as **<2%**, shown via Xen proof-of-construction).